AI Lab
Week

Implement A* search algorithm

8 Puzzle Problem

5	4	
6	1	8
7	3	2

Initial State

	1	2
3	4	5
6	7	8

Goal State

- The 8-puzzle consists of a 3×3 board with eight numbered tiles and a blank space.
- A tile adjacent to the blank space can slide into the space.
- The object is to reach a specified goal state.

A* search algorithm

- **A* search** evaluates nodes by combining g(n), the cost to reach the node and h(n), the cost to get from the node to the goal.
- **f(n) = g(n) + h(n)**
- f(n) is the evaluation function which gives the cheapest solution cost
- g(n) is the exact cost to reach node n from the initial state.
- h(n) is an estimation of the assumed cost from current state (n) to reach the goal.

Heuristic: Number tiles out of place

1. 8-puzzle

2	8	3
1	6	4
	7	5

Initial State n

1	2	3
8		4
7	6	5

Goal State

heuristic(n) = 5

Heuristic: Manhattan Distance

2. 8-puzzle

2	8	3
1	6	4
	7	5

Initial State n

1	2	3
8		4
7	6	5

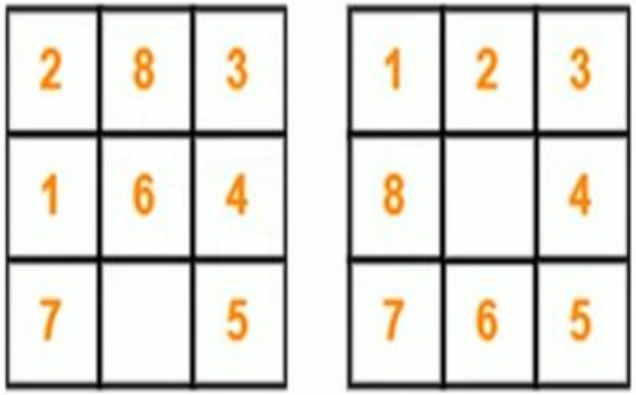
Goal State

heuristic(n) = 1 + 1 + 0 + 0 + 0 + 1 + 1 +8 = 6

Manhattan Distance Heuristic: Another heuristic for 8-puzzle is the Manhattan distance heuristic. This heuristic sums the distance that the tiles are out of place. The distance of a tile is measured by the sum of the differences in the x-positions and the y-positions.

Case1: Implement A* search algorithm

Given an initial state of a 8-puzzle problem and final state to be reached-



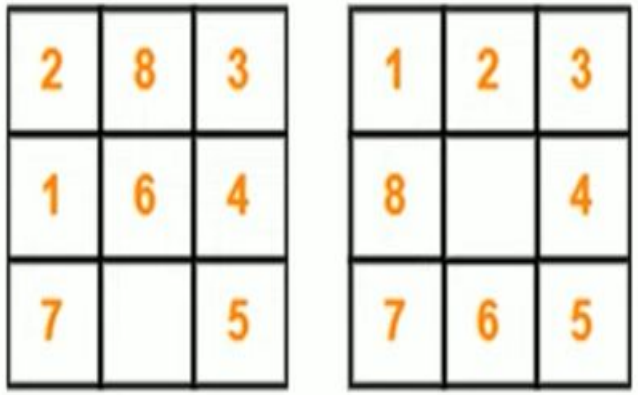
- Find the most cost-effective path to reach the final state from initial state using A* Algorithm.

$$f(n) = g(n) + h(n)$$

- Consider **g(n) = Depth of node** and **h(n) = Number of misplaced tiles**.

Case2: Implement A* search algorithm

Given an initial state of a 8-puzzle problem and final state to be reached-



- Find the most cost-effective path to reach the final state from initial state using A* Algorithm.

$$f(n) = g(n) + h(n)$$

g(n): Depth of the node, h(n): Manhattan Distance